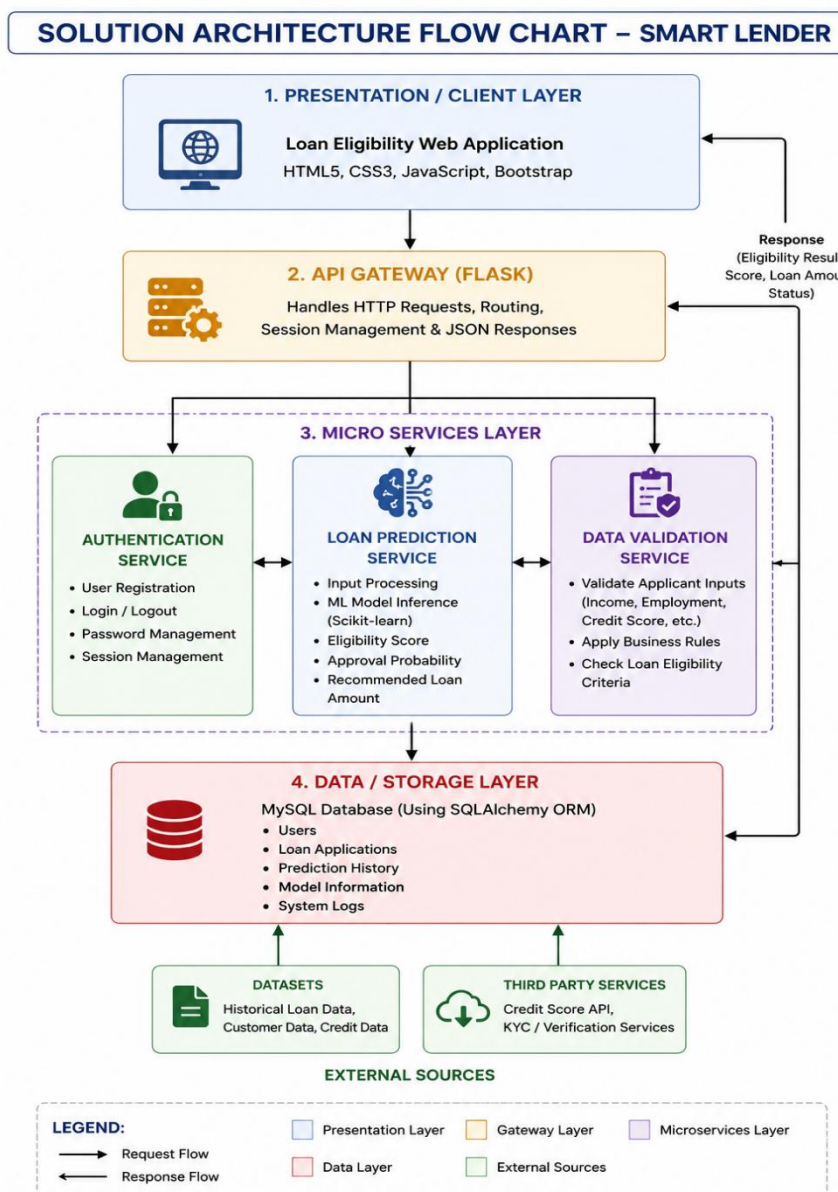


Solution Architecture

Date	30 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

COMPONENT DESCRIPTION TABLE – SMART LENDER		
Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where applicants enter loan details and view loan eligibility predictions.	HTML5, CSS3, JavaScript, Bootstrap
API Gateway	Receives client requests, manages application routing, processes REST APIs, and controls user sessions.	Flask
Authentication Service	Handles user registration, login, authentication, and session management securely.	Python, Flask-Login
Loan Prediction Service	Uses the trained Machine Learning model to analyze applicant details and predict loan eligibility.	Python, Scikit-learn
Data Validation Service	Validates applicant inputs such as income, employment, credit score, and loan amount before prediction.	Python, Flask
Database	Stores user accounts, loan applications, prediction history, and model-related information securely.	MySQL, SQLAlchemy ORM